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(54) **HEAD-WORN DEVICE**

(71) Applicant: **Shenzhen PanBrain Technology Co., Ltd.**, Shenzhen (CN)

(72) Inventor: **Nailin YAO**, Shenzhen (CN)

(73) Assignee: **Shenzhen PanBrain Technology Co., Ltd.**, Shenzhen (CN)

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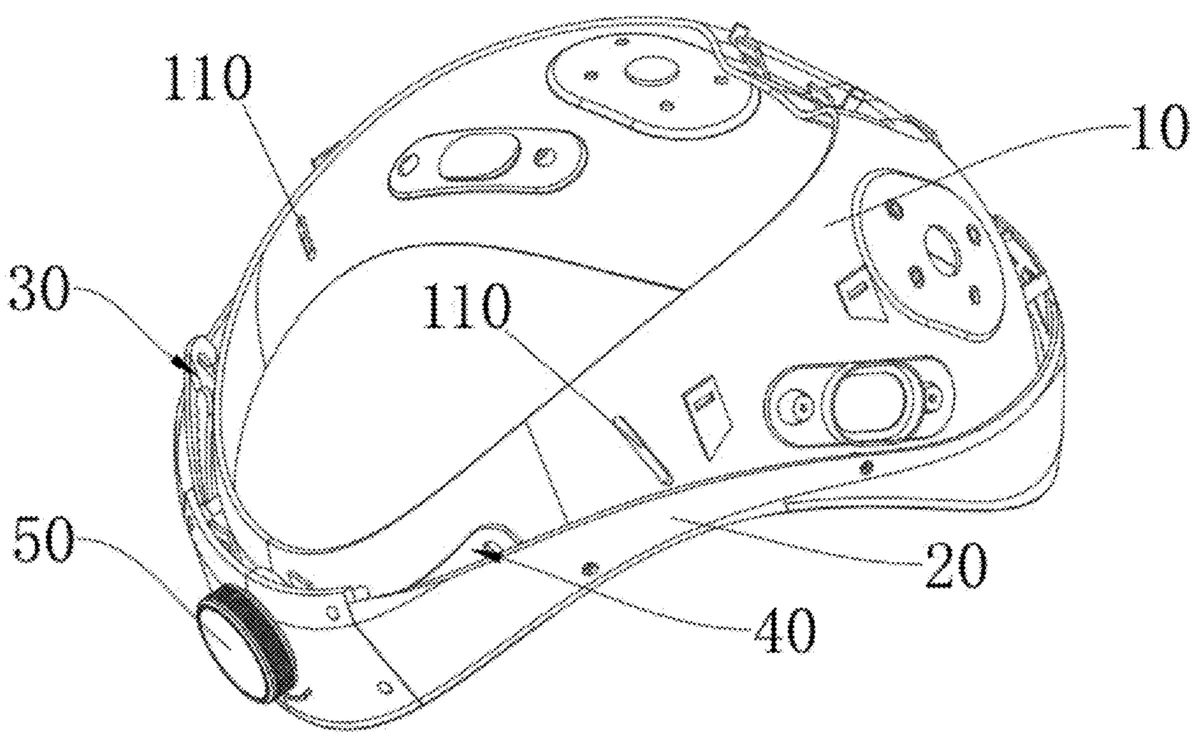
(57) **ABSTRACT**

A head-worn device is arranged with a hollow shell, a first connecting member, a second connecting member, a knob, and a headband. An accommodating space is formed in the hollow shell, and two through holes are arranged on the hollow shell. A mounting hole is arranged on the hollow shell. The mounting hole and the through holes are all connected to the accommodating space. The two connecting members are arranged in the accommodating space, and are able to reciprocate along an axial direction of the accommodating space. The knob is inserted into the mounting hole and extends between the two connecting members. Two sides of the knob are respectively connected to the two connecting members. The headband is arranged on the hollow shell, two ends of the headband respectively pass through the two through holes and are respectively connected to the two connecting members.



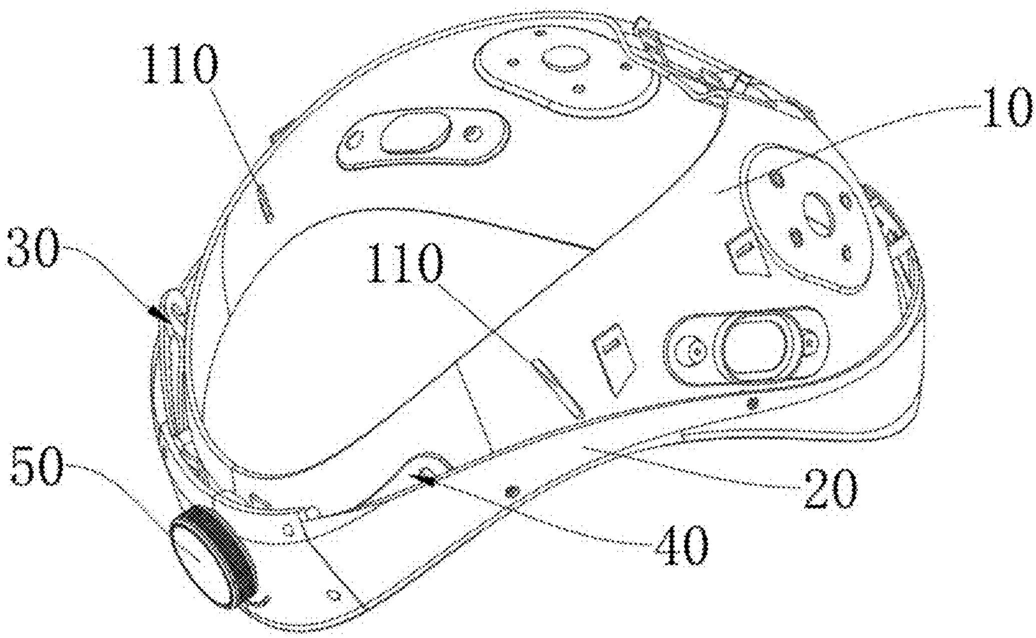


FIG. 1

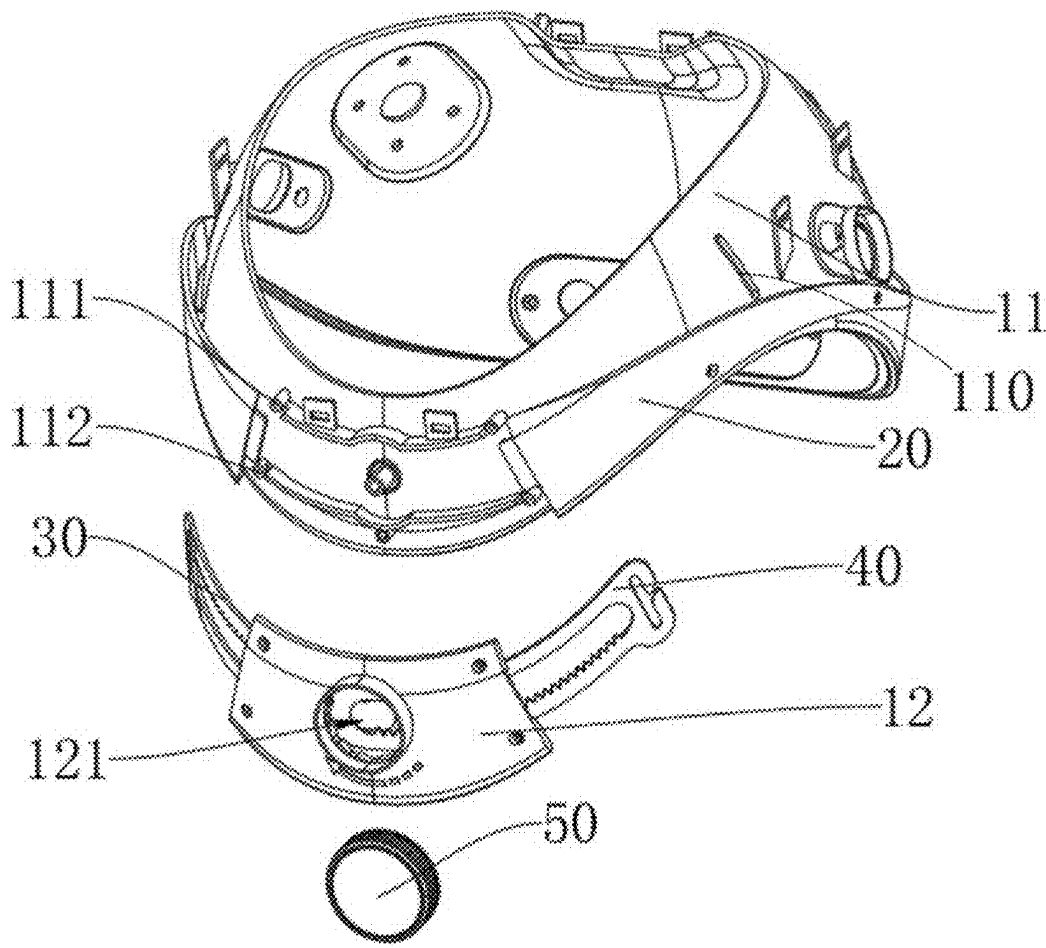


FIG. 2

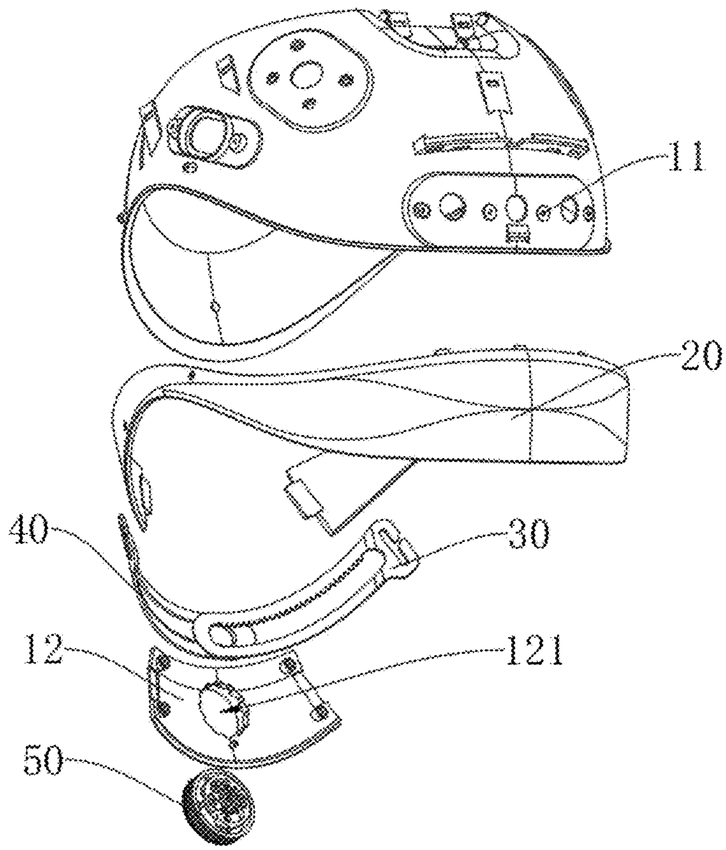

11
20
40
30
12
121
50

FIG. 3

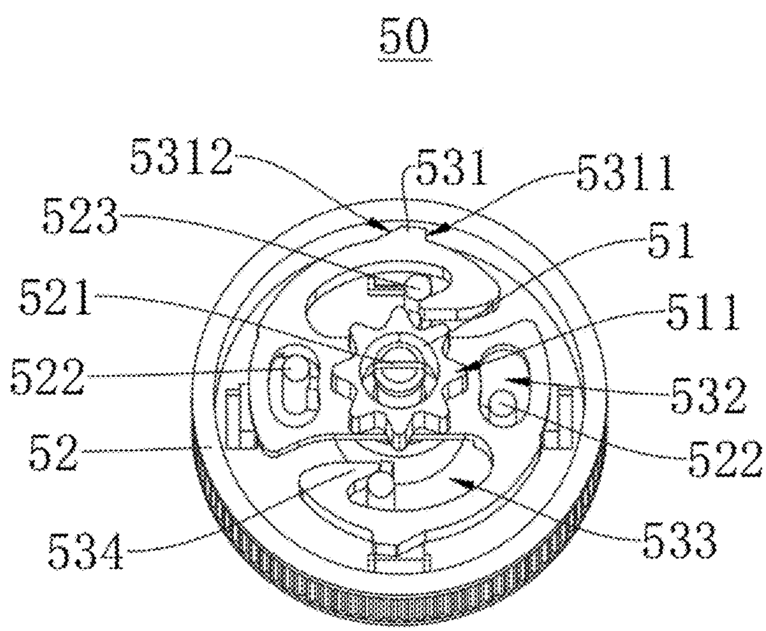

50
5312
531
5311
523
51
521
511
522
532
52
522
534
533

FIG. 4

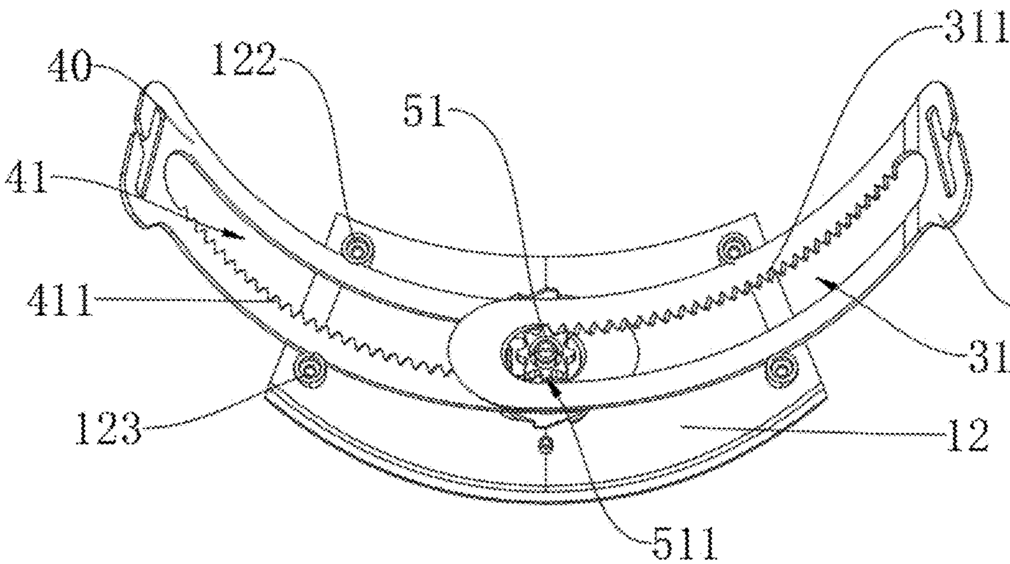

40
122
51
311
41
411
30
31
123
12
511

FIG. 5

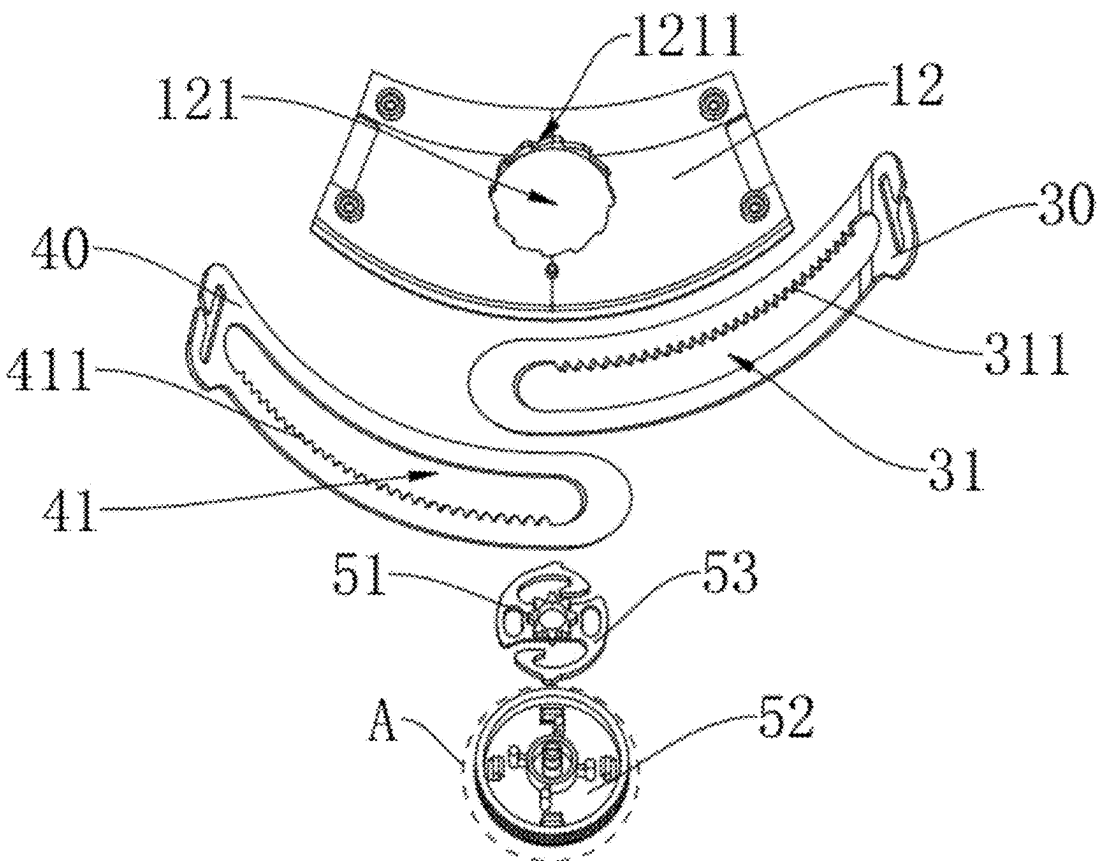

1211
121
12
30
40
311
411
31
41
51
53
A
52

FIG. 6

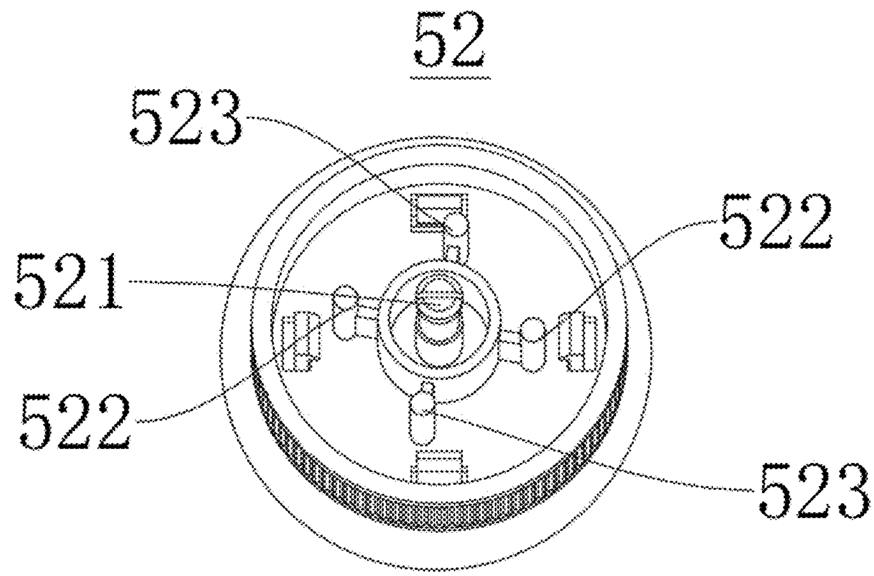


FIG. 7

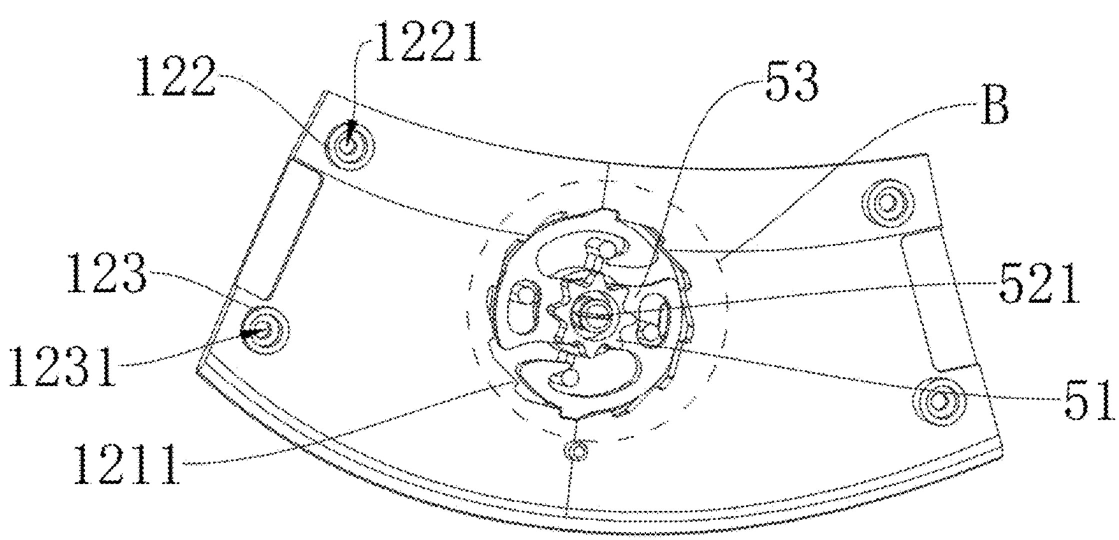


FIG. 8

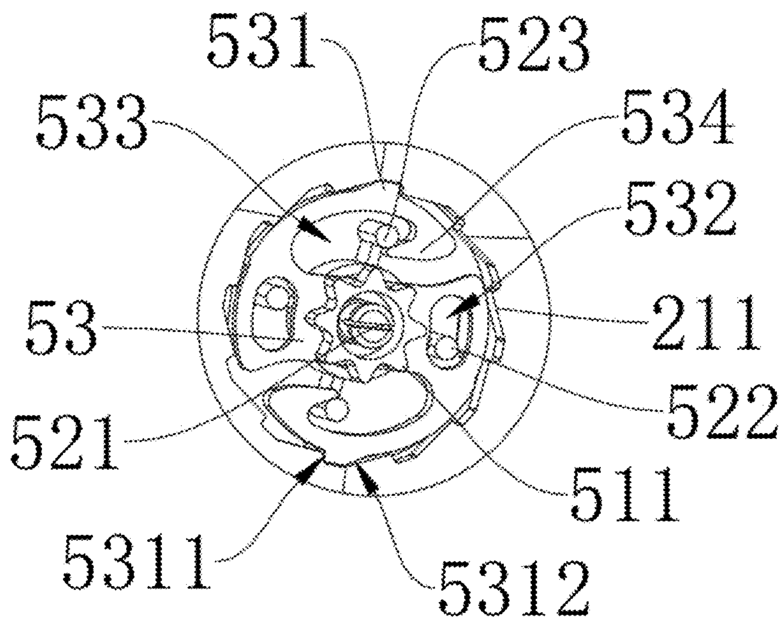


FIG. 9

HEAD-WORN DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present disclosure claims priority to Chinese Patent Application No. 202420473127.4, filed on Mar. 8, 2024, the content of all of which is incorporated herein by reference.

FIELD

[0002] The present disclosure relates to the technical field of wearable devices, in particular to a head-worn device.

BACKGROUND

[0003] With the development of technology, an electroencephalogram is a device used to measure and record human brain activity. The electroencephalogram typically consists of multiple sensors (electrodes) placed on the scalp to capture electrical signals emitted by the brain. These signals are amplified, filtered, and digitized for analysis and interpretation. During the use of a head-mounted electroencephalogram, the device is often worn around the head. However, due to variations in users' head shapes, facial structures, and body sizes, the tightness of the device may differ among different users.

[0004] Accordingly, a reliable and user-friendly adjustment mechanism is needed to regulate the tightness of the head-worn device. A common adjustment method involves the use of an elastic band. For example, Chinese Patent Publication No. CN116036474A discloses a wearable transcutaneous acupoint electrical stimulation therapy device for preventing and treating postoperative cognitive dysfunction. The specification of the patent describes that the therapy device utilizes an adjustable elastic band along the coronal plane of the head to accommodate different head circumferences and securely fasten the wearable device to the patient's head.

[0005] However, elastic band generally provides only limited adjustability in terms of tightness and tends to lose elasticity over prolonged use, making effective adjustment of tightness difficult.

[0006] Therefore, the prior art still needs to be improved and developed.

SUMMARY

[0007] In view of the above-mentioned deficiencies in the prior art, an object of the present disclosure is to provide a head-worn device, aiming to solve the problem that the elastic band can only provide limited tightness adjustment and loses elasticity during prolonged use, resulting in difficulties of effective adjustment of tightness.

[0008] The technical solution of the present disclosure is as follows:

[0009] A head-worn device includes a hollow shell, a first connecting member, a second connecting member, a knob, and a headband, an accommodating space is formed in the hollow shell, and two through holes are oppositely arranged on an inner side wall of the hollow shell; a mounting hole is arranged on an outer side wall of the hollow shell; the mounting hole and the through holes are all connected to the accommodating space; the first connecting member and the second connecting member are arranged in parallel in the accommodating space, and the first connecting member and the second connecting member are both able to reciprocate along an axial direction of the accommodating space; the knob is inserted into the mounting hole and extends between the first connecting member and the second connecting member; two sides of the knob are respectively connected to the first connecting member and the second connecting member, and are used to drive the first connecting member and the second connecting member to move toward or away from each other; and the headband is arranged on a side wall of the hollow shell; two ends of the headband respectively pass through the two through holes and are respectively connected to the first connecting member and the second connecting member.

[0010] In the head-worn device, a first strip-shaped through hole is arranged on the first connecting member, and a first rack is arranged on a side wall of the first strip-shaped through hole along a length direction; a second strip-shaped through hole is arranged on the second connecting member, and a second rack is arranged on a side wall of the second strip-shaped through hole along a length direction; the first rack and the second rack are arranged opposite to each other, and the knob includes an adjusting gear, the adjusting gear is rotatably arranged in the first strip-shaped through hole and the second strip-shaped through hole, and a gear ring portion is arranged on a circumference of the adjusting gear, and the first rack and the second rack are respectively meshed with opposite sides of the gear ring portion.

[0011] In the head-worn device, a ratchet ring is arranged on a side wall of the mounting hole, and the knob includes a mounting cover and a stopping body; the mounting cover is arranged on one side of the mounting hole away from the accommodating space, and a central shaft is arranged on the mounting cover, and the adjusting gear is sleeved on the central shaft, the mounting cover is used to drive the adjusting gear to rotate; the stopping body is sleeved on the central shaft and rotates synchronously with the adjusting gear, a pawl portion is arranged on an outer peripheral wall of the stopping body, and the pawl portion includes a stopping surface and a guiding inclined surface, and when the stopping body rotates in a direction toward the stopping surface, the stopping surface engages with the ratchet ring; when the stopping body rotates in a direction toward the guiding inclined surface, the guiding inclined surface and the ratchet ring slide relative to each other.

[0012] In the head-worn device, an angle between the guiding inclined surface and the stopping body is an acute angle; and an angle between the stopping surface and the stopping body is a right angle or an obtuse angle.

[0013] In the head-worn device, a rib column is arranged on a side of the mounting cover facing the accommodating space, and an arc groove adapted to the rib column is arranged on the stopping body, the rib column is inserted into the arc groove, and when the mounting cover rotates, the rib column abuts against the arc groove to push the stopping body to rotate.

[0014] In the head-worn device, a push column is further arranged on the side of the mounting cover facing the accommodating space, a hook opening is arranged on the stopping body, an end of the stopping body is bent toward the hook opening to form a hook-shaped part, and the push column is arranged to pass through the hook opening, when the stopping body rotates in a direction toward the stopping surface, the push column pushes the hook-shaped part inward to disengage the stopping surface and the ratchet ring.

[0015] In the head-worn device, the stopping body and the adjusting gear are integrally formed; and/or the stopping body and the pawl portion are integrally formed.

[0016] In the head-worn device, at least two pawl portions are arranged at intervals along a circumferential direction on an outer edge of the ratchet ring.

[0017] In the head-worn device, the hollow shell includes a wearable shell and an assembly shell, the assembly shell cooperates with the wearable shell to form the accommodating space, the headband is arranged on the wearable shell, and the mounting hole is opened on the assembly shell; a first guiding column and a second guiding column are arranged on a side of the assembly shell facing the accommodating space, the first guiding column and the second guiding column are arranged at intervals along a width direction of the assembly shell, and the first connecting member or the second connecting member is arranged between the first guiding column and the second guiding column.

[0018] In the head-worn device, a first positioning groove is arranged on the first guiding column, a first positioning column adapted to the first positioning groove is arranged on the wearable shell, and the first positioning column is inserted into the first positioning groove; and/or a second positioning groove is arranged on the second guiding column, a second positioning column adapted to the second positioning groove is arranged on the wearable shell, and the second positioning column is inserted into the second positioning groove.

[0019] Compared with the prior art, the embodiment of the present disclosure has the following advantages:

[0020] The present disclosure provides a head-worn device with adjustable tightness and easy operation, which is arranged with a hollow shell for wearing on the top of the user's head, and is also arranged with a first connecting member, a second connecting member and a headband. The headband passes through through holes and is connected with the first connecting member and the second connecting member, enabling the entire head-mounted device to be securely fastened to the user's head, thereby reducing movement and shaking. The first connecting member and the second connecting member, connected with the headband, form an annular structure that encircles the user's head. The user can adjust the relative movement of the first connecting member and the second connecting member toward or away from each other via a knob, thereby adjusting the length of the overlapping portion between the first connecting member and the second connecting member. The adjustment can be performed without additional tools or operations, allowing for precise tightness adjustment to accommodate users with different head sizes. Furthermore, the adjustment can be performed by touch without requiring visual confirmation, making the operation simple and convenient. The present disclosure reduces sliding and shaking of the head-worn device, providing improved stability.

[0021] The head-worn device can still maintain stability and comfort during long-term use. By using the head-worn device, the tightness of the head-worn device can be flexibly adjusted according to the head circumference of different users to obtain better wearing comfort and stability.

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] In order to more clearly illustrate the embodiments of the present disclosure or the technical solutions in the prior art, the drawings required for use in the embodiments are briefly introduced below. Obviously, the drawings described below are only some embodiments recorded in the present disclosure. For those skilled in the art, other drawings can be obtained based on these drawings without involving creative work.

[0023] FIG. **1** is an assembly diagram of a head-worn device according to the present disclosure.

[0024] FIG. **2** is an exploded view of the structure of the head-worn device according to the present disclosure.

[0025] FIG. **3** is another exploded view of the structure of the head-worn device according to the present disclosure.

[0026] FIG. **4** is a schematic diagram of the structure of the knob according to the present disclosure.

[0027] FIG. **5** is an assembly diagram of the knob, the first connecting member, the second connecting member and the assembly shell according to the present disclosure.

[0028] FIG. **6** is an exploded view of the structure of the knob, the first connecting member, the second connecting member and the assembly shell according to the present disclosure.

[0029] FIG. **7** is an enlarged view of part A in FIG. **6**.

[0030] FIG. **8** is a schematic diagram of the structure of the assembly shell and the knob according to the present disclosure.

[0031] FIG. **9** is an enlarged view of part B in FIG. **8**.

DETAILED DESCRIPTION OF EMBODIMENTS

[0032] In order to enable those skilled in the art to better understand the technical solution of the present disclosure, the drawings of the embodiments of the present disclosure are combined to clearly and completely describe the technical solutions in the embodiments of the present disclosure. Obviously, the described embodiments are only part of the embodiments of the present disclosure, not all of the embodiments. Based on the embodiments of the present disclosure, all other embodiments obtained by those skilled in the art without creative work are within the scope of protection of the present disclosure.

[0033] Referring to FIG. **1**, FIG. **2** and FIG. **3**, in one embodiment of the present disclosure, a head-worn device is disclosed, which includes a hollow shell **10**, a first connecting member **30**, a second connecting member **40**, a knob **50**, and a headband **20**. An accommodating space is formed in the hollow shell **10**, and two through holes **110** are oppositely arranged on an inner side wall of the hollow shell **10**. A mounting hole **121** is arranged on an outer side wall of the hollow shell **10**; the mounting hole **121** and the through holes **110** are all connected to the accommodating space. The first connecting member **30** and the second connecting member **40** are arranged in parallel in the accommodating space, and the first connecting member **30** and the second connecting member **40** can reciprocate along an axial direction of the accommodating space. The knob **50** is inserted into the mounting hole **121** and extends between the first connecting member **30** and the second connecting member **40**. Two sides of the knob **50** are respectively connected to the first connecting member **30** and the second connecting member **40**, and are used to drive the first connecting member **30** and the second connecting member **40** to move toward or away from each other. The headband **20** is arranged on a side wall of the hollow shell **10**. Two ends of the headband **20** respectively pass through the two through holes **110**, and are respectively connected to the first connecting member **30** and the second connecting member **40**.

[0034] The present embodiment provides a head-worn device with adjustable tightness and easy operation, which is arranged with a hollow shell **10** for wearing on the top of the user's head, and is also arranged with a first connecting member **30**, a second connecting member **40** and a headband **20**. The headband **20** passes through through holes **110** and is connected with the first connecting member **30** and the second connecting member **40**, enabling the entire head-mounted device to be securely fastened to the user's head, thereby reducing movement and shaking. The first connecting member **30** and the second connecting member **40**, together with the headband **20**, form an annular structure that encircles the user's head. The user can adjust the relative movement of the first connecting member **30** and the second connecting member **40** toward or away from each other via a knob **50**, thereby adjusting the length of the overlapping portion between the first connecting member **30** and the second connecting member **40**. The adjustment can be performed without additional tools or operations, allowing for precise tightness adjustment to accommodate users with different head sizes. Furthermore, the adjustment can be performed by touch without requiring visual confirmation, making the operation simple and convenient. The present disclosure reduces sliding and shaking of the head-worn device, providing improved stability

[0035] Exemplarily, after the user puts on the head-worn device, if the user feels that the head and the head are not firmly fit, the knob **50** can be used to adjust the first connecting member **30** and the second connecting member **40** to move to each other, that is, the length of the overlapping portion between the first connecting member **30** and the second connecting member **40** increases, so that the enclosure structure in contact with the head is tightened as a whole. If the user feels too much pressure on the head after putting on the device, that is, the enclosure is too tight, the knob **50** can be used to adjust the first connecting member **30** and the second connecting member **40** to move away from each other, that is, the length of the overlapping portion between the first connecting member **30** and the second connecting member **40** decreases, so that the enclosure structure in contact with the head becomes loose as a whole and is adjusted to a state suitable for the head circumference.

[0036] The disclosed head-worn device can still maintain stability and comfort during long-term use. By using the disclosed head-worn device, the tightness of the head-worn device can be flexibly adjusted according to the head circumference of different users to obtain better wearing comfort and stability.

[0037] As shown in FIG. **5** and FIG. **6**, as another implementation of the present embodiment, the first connecting member **30** is disclosed to be arranged with a first strip-shaped through hole **31**, and a first rack **311** is arranged on a side wall of the first strip-shaped through hole **31** along a length direction. The second connecting member **40** is arranged with a second strip-shaped through hole **41**, and a second rack **411** is arranged on a side wall of the second strip-shaped through hole **41** along a length direction, the first rack **311** and the second rack **411** are arranged opposite to each other. The knob **50** includes an adjusting gear **51**, and the adjusting gear **51** is rotatably arranged in the first strip-shaped through hole **31** and the second strip-shaped through hole **41**. A gear ring portion **511** is arranged on a circumference of the adjusting gear **51**, the first rack **311** and the second rack **411** are respectively meshed with opposite sides of the gear ring portion **511**. The adjusting gear **51** is used to drive the first connecting member **30** and the second connecting member **40** move toward or away from each other, and the tightness of the head circumference can be adjusted by rotating the adjusting gear **51**, which is easy to operate and convenient and quick.

[0038] The length of the first rack **311** and the second rack **411** disclosed in the present embodiment is the stroke length that the adjusting gear **51** is able to adjust. The adjusting gear **51** meshes the first rack **311** and the second rack **411** to increase or decrease the length of the overlapping portions. Here, the length of the first rack **311** and the second rack **411** is not limited and can be set according to actual usage requirements.

[0039] The first connecting member **30** and the second connecting member **40** disclosed in the present embodiment are made of PP material, which has excellent softness and plastic strength. The first connecting member **30** and the second connecting member **40** can move toward each other while driven by the adjusting gear **51** to decrease the head circumference size, or move away from each other while driven by the adjusting gear **51** to increase the head circumference size.

[0040] It should be noted that, in the present embodiment, only the materials of the first connecting member **30** and the second connecting member **40** are exemplified, but the protection scope of the present disclosure is not limited thereto. Other types of materials that are able to achieve the technical effects disclosed in the present disclosure and are equivalent replacements for the concept of the present disclosure, should also be within the scope of protection of the present disclosure.

[0041] The first connecting member **30** and the second connecting member **40** disclosed in the present embodiment are arranged in an arc shape, and two free ends are connected to the head-worn device to form a ring shape surrounding the user's head. The first connecting member **30** and the second connecting member **40** are also arranged with hollow grooves, and the hollow grooves can reduce the weight of the first connecting member **30** and the second connecting member **40**, thereby reducing the difficulty of wearing.

[0042] The first rack **311** and the second rack **411** disclosed in the present embodiment are arranged opposite to each other. In the present embodiment, as shown in FIG. **2**, the first rack **311** is arranged on an upper side of the first strip-shaped through hole **31** and on a left side of the adjusting gear **51**, and the second rack **411** is arranged on a lower side of the second strip-shaped through hole **41** and on a right side of the adjusting gear **51**. The two racks are arranged relative to each other and mesh with the adjusting gear **51**, that is, when the adjusting gear **51** rotates clockwise, the first connecting member **30** and the second connecting member **40** move toward each other; when the adjusting gear **51** rotates counterclockwise, the first connecting member **30** and the second connecting member **40** move away from each other to achieve tightness adjustment.

[0043] As shown in FIG. **4**, FIG. **5** and FIG. **6**, as another implementation of the present embodiment, a ratchet ring **1211** is disclosed to be arranged on a side wall of the mounting hole **121**, and the knob **50** includes a mounting cover **52**, which is arranged on a side of the mounting hole **121** away from the accommodating space. That is, the mounting cover **52** is arranged on the outside for the convenience of the user to touch and rotate. A central shaft **521** is arranged on a side of the mounting cover **52** facing the accommodating space, and the adjusting gear **51** is sleeved on the central shaft **521**. The mounting cover **52** is used to drive the adjusting gear **51** to rotate, and through the mounting cover **52** the user can drive the central shaft **521** and the adjusting gear **51** sleeved on the central shaft **521** to rotate, so as to flexibly adjust the movement of the first connecting member **30** and the second connecting member **40**.

[0044] The adjusting gear **51** disclosed in the present embodiment is fixedly connected to the mounting cover **52**. The central shaft **521** is arranged with a support column protruding along its radial direction, the adjusting gear **51** is arranged with a central hole for inserting the central shaft **521**, and a side wall of the central hole is arranged with a support groove adapted to the support column. The adjusting gear **51** and the mounting cover **52** are fixedly connected via the support column and the support groove. It should be noted that the fixed connection method between the adjusting gear **51** and the central shaft **521** is not limited to the present embodiment.

[0045] The knob **50** disclosed in the present embodiment further includes a stopping body **53**, which is sleeved on the central shaft **521** and rotates synchronously with the adjusting gear **51**. A pawl portion **531** is arranged on an outer peripheral wall of the stopping body **53**. The pawl portion **531** includes a stopping surface **5311** and a guiding inclined surface **5312**. When the stopping body **53** rotates in a direction toward the stopping surface **5311**, the stopping surface **5311** engages with the ratchet ring **1211**; when the stopping body **53** rotates in a direction toward the guiding inclined surface **5312**, the guiding inclined surface **5312** and the ratchet ring **1211** slide relative to each other.

[0046] The stopping surface **5311** of the pawl portion **531** disclosed in the present embodiment engages with the ratchet ring **1211** to play a certain buffering role, which can limit the rotation of the central shaft **521**, so that the adjusting gear **51** stays at a specific position. The user can conveniently fix the mounting cover **52** at a position with desired tightness to avoid the adjustment of the adjusting gear **51** due to accidental rotation during use, and prevent the first connecting member **30** and the second connecting member **40** from loosening by themselves. Even if the head-worn device is affected by movement or external force, the first connecting member **30** and the second connecting member **40** can be stably maintained at the position with desired tightness and are not prone to loosening or sliding.

[0047] The guiding inclined surface **5312** disclosed in the present embodiment is design to slide relative to the ratchet ring **1211** when the first connecting member **30** and the second connecting member **40** move toward each other, while the stopping surface **5311** and the ratchet ring **1211** are designed to be engaged with each other when the first connecting member **30** and the second connecting member **40** move away from each other, since the first connecting member **30** and the second connecting member **40** may loosen themselves, and can not tighten by themselves.

[0048] As another implementation of the present embodiment, an angle between the guiding inclined surface **5312** and the stopping body **53** is an acute angle. When the stopping body rotates toward the guiding inclined surface **5312**, the angle between the guiding inclined surface **5312** and the stopping body being an acute angle would not lead to contact surfaces against each other. The ratchet ring **1211** does not limit the rotation of the stopping body **53** toward the guiding inclined surface **5312**. An angle between the stopping surface **5311** and the stopping body **53** is a right angle or an obtuse angle. When the stopping body rotates toward the stopping surface **5311**, contact surfaces between the stopping surface **5311** and the ratchet ring **1211** against each other, so that the pawl portion **531** is engaged with the ratchet ring **1211**, thereby limiting the rotation of the stopping body **53**.

[0049] Exemplarily, as shown in FIG. **5**, when the first connecting member **30** and the second connecting member **40** loosen by themselves, the adjusting gear **51** is driven to rotate clockwise. At this time, due to the engagement of the stopping surface **5311** and the ratchet ring **1211**, the adjusting gear **51** is prevented from rotating, that is, the first connecting member **30** and the second connecting member **40** are prevented from loosening by themselves.

[0050] As another implementation of the present embodiment, the stopping body **53** and the adjusting gear **51** are integrally formed. When the user adjusts the mounting cover **52**, the stopping body **53** and the adjusting gear **51** rotate synchronously to more accurately adjust the tightness of the head circumference. The connection between the stopping body **53** and the adjusting gear **51** is firm, which reduces the possibility of loosening and swinging, and improves the assembly quality and service life.

[0051] The stopping body **53** disclosed in the present embodiment is integrally formed with the pawl portion **531**, so that when the pawl portion **531** is engaged with the ratchet ring **1211**, the pawl portion **531** is difficult to be damaged, thereby extending the service life of the pawl portion **531**.

[0052] As another implementation of the present embodiment, at least two pawl portions **531** are arranged at intervals along a circumferential direction on an outer edge of the ratchet ring **1211**. The arrangement of the two pawl portions **531** can provide better connection stability, increase the abutment area, and evenly share the bearing force of the connecting members. The two pawl portions **531** being arranged at intervals along circumferential direction can better resist the influence of external forces and reduce the risk of loosening of the first connecting member **30** and the second connecting member **40**. The head-worn device is ensured to remain stable during use and is not prone to loosening or falling off.

[0053] As shown in FIG. **7**, FIG. **8** and FIG. **9**, as another implementation of the present embodiment, a rib column **522** is arranged on one side of the mounting cover **52** facing the accommodating space, and an arc groove **532** adapted to the rib column **522** is arranged on the stopping body **53**. The rib column **522** is inserted into the arc groove **532**. When the mounting cover **52** rotates, the rib column **522** abuts against the arc groove **532** to push the stopping body **53** to rotate. The purpose of providing the rib column **522** is to enable the stopping body **53** to rotate when adjusting the mounting cover **52**, and also to increase the connection mode between the mounting cover **52** and the stopping body **53**, so that the structure is more compact and the connection is more firm.

[0054] Two rib columns **522** and two arc grooves **532** are disclosed in the present embodiment and the two rib columns **522** and the two arc grooves **532** are arranged correspond to each other. Through the positioning of the two rib columns **522** and the guidance of the two arc grooves **532**, the accurate alignment of the components during the assembly process can be ensured. During the rotation of the mounting cover **52**, the two rib columns **522** contact an inner wall of the arc grooves **532**, pushing the stopping body **53** to rotate.

[0055] As another implementation of the present embodiment, a push column **523** is further arranged on the side of the mounting cover **52** facing the accommodating space, and a hook opening **533** is arranged on the stopping body **53**. An end of the stopping body **53** is bent toward the hook opening **533** to form a hook-shaped part **534**, and the push column **523** is inserted into the hook opening **533**. When the stopping body **53** rotates in a direction toward the stopping surface **5311**, the push column **523** pushes the hook-shaped part **534** inward to ensure the stopping surface **5311** is disengaged with the ratchet ring **1211**.

[0056] Exemplarily, as shown in FIG. **9**, when the stopping body **53** rotates in the clockwise direction, the push column **523** moves in a direction toward the hook-shaped part **534** and pushes the hook-shaped part **534**, so that the hook-shaped part **534** is deformed toward an axis of the stopping body **53**, thereby driving the pawl portion **531** on the stopping body **53** to also deform toward the axis of the stopping body **53**, so that the pawl portion **531** is no longer engaged with the ratchet ring **1211** and is out of engagement. At this time, the first connecting member **30** and the second connecting member **40** can be adjusted to move toward or away from each other at will. When the mounting cover **52** is rotated to a suitable position, the rotational force is no longer applied to the stopping body **53**, the push column **523** no longer pushes the hook-shaped part **534**, and the pawl portion **531** resumes to be engaged with the ratchet ring **1211**.

[0057] When rotating counterclockwise, the contact surface between the pawl portion **531** and the ratchet ring **1211** is an inclined surface, and the pawl portion **531** does not produce a blocking effect. Therefore, the length of the overlapping portion can be easily adjusted by rotating the mounting cover **52** with one hand, which is easy to operate.

[0058] Two push columns **523** and two hook openings **533** are disclosed in the present embodiment, which increase the reliability of the head-worn device. The hook-shaped part **534** and the pawl portion **531** are arranged on a same radial direction of the stopping body, so that when the hook-shaped part **534** is deformed, the pawl portion **531** is also deformed synchronously to release the engaging state.

[0059] The push columns **523** and the rib columns **522** disclosed in the present embodiment are arranged in an alternating sequence at intervals. The arrangement of the push columns **523** can also reduce the abutment force between the pawl portion **531** and the ratchet ring **1211**, prevent the pawl portion **531** from breaking, and extend the service life of the pawl portion **531**.

[0060] The stopping body **53** disclosed in the present embodiment is an elastic stopping body **53**. When the pawl portion **531** is subjected to a large rotational force, the pawl portion **531** can produce elastic deformation to prevent the pawl portion **531** from breaking. By setting the knob **50**, the number of hardware is reduced, which can reduce the weight of the head-worn device.

[0061] As shown in FIG. **5** and FIG. **8**, as another implementation of the present embodiment, the hollow shell **10** includes a wearable shell **11** and an assembly shell **12**, the assembly shell **12** cooperates with the wearable shell **11** to form the accommodating space. The headband **20** is arranged on a side wall of the wearable shell **11**, and the mounting hole **121** is opened on the assembly shell **12**. A first guiding column **122** and a second guiding column **123** are arranged on a side of the assembly shell **12** facing the accommodating space, the first guiding column **122** and the second guiding column **123** are arranged at intervals along a width direction of the assembly shell **12**, and the first connecting member **30** or the second connecting member **40** is arranged between the first guiding column **122** and the second guiding column **123**, that is, the first connecting member **30** or the second connecting member **40** is clamped between the first guiding column **122** and the second guiding column **123**. Therefore, the first connecting member **30** and the second connecting member **40** are facilitated to move along a length direction of the assembly shell **12**, thereby improving the efficiency of the first connecting member **30** and the second connecting member **40** moving toward or away from each other.

[0062] Further, a first positioning groove **1221** is arranged on the first guiding column **122** disclosed in the present embodiment, and a first positioning column **111** adapted to the first positioning groove **1221** is arranged on the wearable shell **11**. The first positioning column **111** is inserted into the first positioning groove **1221**. And/or, a second positioning groove **1231** is arranged on the second guiding column **123**, and a second positioning column **112** adapted to the second positioning groove **1231** is arranged on the wearable shell **11**. The second positioning column **112** is inserted into the second positioning groove **1231**. Thus, the wearable shell **11** is fixedly connected to the assembly shell **12**, thereby improving the overall reliability.

[0063] In summary, the present disclosure provides a head-worn device with adjustable tightness and easy operation, which is arranged with a hollow shell **10** for wearing on the top of the user's head, and is also arranged with a first connecting member **30**, a second connecting member **40** and a headband **20**. The headband **20** passes through through holes **110** and is connected to the first connecting member **30** and the second connecting member **40**, so that the entire head-worn device can be tightly fixed on the user's head to reduce movement and shaking. The first connecting member **30** and the second connecting member **40**, connected with the headband **20**, form an annular structure that encircles the user's head. The user can adjust the relative movement of the first connecting member **30** and the second connecting member **40** toward or away from each other via a knob **50**, thereby adjusting the length of the overlapping portion between the first connecting member **30** and the second connecting member **40**. The adjustment can be performed without additional tools or operations, allowing for precise tightness adjustment to accommodate users with different head sizes. Furthermore, the adjustment can be performed by touch without requiring visual confirmation, making the operation simple and convenient. The present disclosure reduces sliding and shaking of the head-worn device, providing improved stability.

[0064] The disclosed head-worn device provides stability and comfort during long-term use. By using the head-worn device, the tightness of the head-worn device can be flexibly adjusted according to the head circumference of different users to obtain better wearing comfort and stability.

[0065] It should be noted that, in the absence of conflict, the embodiments and features in the embodiments of the present disclosure may be combined with each other.

[0066] It should be noted that the present disclosure takes a head-worn device as an example to introduce the structure and working principle of the present disclosure, but the present disclosure is not limited to the head-worn device, and can also be applied to the production and use of other similar products.

[0067] It should be understood that the present disclosure is not limited to the precise structure described above and shown in the drawings, and various modifications and substitutions can be made without departing from the protecting scope. The scope of the present disclosure is limited only by the appended claims.

[0068] The above description is only an embodiment of the present disclosure and is not intended to limit the present disclosure. Any modifications, equivalent substitutions, improvements, etc. made within the spirit and principles of the present disclosure shall be included in the protection scope of the present disclosure.

What is claimed is:

1. A head-worn device, comprising:

a hollow shell, an accommodating space is formed in the hollow shell, and two through holes are oppositely arranged on an inner side wall of the hollow shell; a mounting hole is arranged on an outer side wall of the hollow shell; wherein the mounting hole and the through holes are all connected to the accommodating space;

a first connecting member and a second connecting member, the first connecting member and the second connecting member are arranged in parallel in the accommodating space, and the first connecting member and the second connecting member are both able to reciprocate along an axial direction of the accommodating space;

a knob, the knob is inserted into the mounting hole and extends between the first connecting member and the second connecting member; two sides of the knob are respectively connected to the first connecting member and the second connecting member, and are configured to drive the first connecting member and the second connecting member to move toward or away from each other; and

a headband, the headband is arranged on a side wall of the hollow shell; two ends of the headband respectively pass through the two through holes and are respectively connected to the first connecting member and the second connecting member.

2. The head-worn device according to claim **1**, wherein a first strip-shaped through hole is arranged on the first connecting member, and a first rack is arranged on a side wall of the first strip-shaped through hole along a length direction; a second strip-shaped through hole is arranged on the second connecting member, and a second rack is arranged on a side wall of the second strip-shaped through hole along a length direction;

wherein the first rack and the second rack are arranged opposite to each other, and the knob comprises an adjusting gear, the adjusting gear is rotatably arranged in the first strip-shaped through hole and the second strip-shaped through hole, a gear ring portion is arranged on a circumference of the adjusting gear, and the first rack and the second rack are respectively meshed with opposite sides of the gear ring portion.

3. The head-worn device according to claim **2**, wherein a ratchet ring is arranged on a side wall of the mounting hole, and the knob comprises:

a mounting cover, the mounting cover is arranged on one side of the mounting hole away from the accommodating space, a central shaft is arranged on the mounting cover, and the adjusting gear is sleeved on the central shaft, the mounting cover is configured to drive the adjusting gear to rotate; and

a stopping body, the stopping body is sleeved on the central shaft and rotates synchronously with the adjusting gear, a pawl portion is arranged on an outer peripheral wall of the stopping body, the pawl portion comprises a stopping surface and a guiding inclined surface, and when the stopping body rotates in a direction toward the stopping surface, the stopping surface engages with the ratchet ring; when the stopping body rotates in a direction toward the guiding inclined surface, the guiding inclined surface and the ratchet ring slide relative to each other.

4. The head-worn device according to claim **3**, wherein an angle between the guiding inclined surface and the stopping body is an acute angle; and

an angle between the stopping surface and the stopping body is a right angle or an obtuse angle.

5. The head-worn device according to claim **3**, wherein a rib column is arranged on a side of the mounting cover facing the accommodating space, and an arc groove adapted to the rib column is arranged on the stopping body, the rib column is inserted into the arc groove, and when the mounting cover rotates, the rib column abuts against the arc groove to push the stopping body to rotate.

6. The head-worn device according to claim **3**, wherein a push column is arranged on a side of the mounting cover facing the accommodating space, a hook opening is arranged on the stopping body, an end of the stopping body is bent toward the hook opening to form a hook-shaped part, and the push column is arranged to pass through the hook opening, when the stopping body rotates in a direction toward the stopping surface, the push column pushes the hook-shaped part inward to disengage the stopping surface and the ratchet ring.

7. The head-worn device according to claim **3**, wherein the stopping body and the adjusting gear are integrally formed; and/or

the stopping body and the pawl portion are integrally formed.

8. The head-worn device according to claim **4**, wherein at least two pawl portions are arranged at intervals along a circumferential direction on an outer edge of the ratchet ring.

9. The head-worn device according to claim **1**, wherein the hollow shell comprises a wearable shell and an assembly shell, the assembly shell cooperates with the wearable shell to form the accommodating space, the headband is arranged on the wearable shell, and the mounting hole is opened on the assembly shell;

wherein a first guiding column and a second guiding column are arranged on a side of the assembly shell facing the accommodating space, the first guiding column and the second guiding column are arranged at intervals along a width direction of the assembly shell, and the first connecting member or the second connecting member is arranged between the first guiding column and the second guiding column.

10. The head-worn device according to claim **9**, wherein a first positioning groove is arranged on the first guiding column, a first positioning column adapted to the first positioning groove is arranged on the wearable shell, and the first positioning column is inserted into the first positioning groove; and/or

a second positioning groove is arranged on the second guiding column, a second positioning column adapted to the second positioning groove is arranged on the wearable shell, and the second positioning column is inserted into the second positioning groove.

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